*For the use only of a Registered Medical Practitioner*

PRESCRIBING INFORMATION

# ENHANCIN TABLETS

(Amoxicillin and Clavulanate Potassium Tablets)

To reduce the development of drug-resistant bacteria and maintain the effectiveness of amoxicillin/clavulanate potassium and other antibacterial drugs, amoxicillin/clavulanate potassium should be used only to treat or prevent infections that are proven or strongly suspected to be caused by bacteria.

## COMPOSITION

Each tablet contains:
Amoxicillin trihydrate Ph. Eur. equivalent to Amoxicillin…500 mg
Potassium Clavulanate equivalent to Clavulanic acid….125 mg

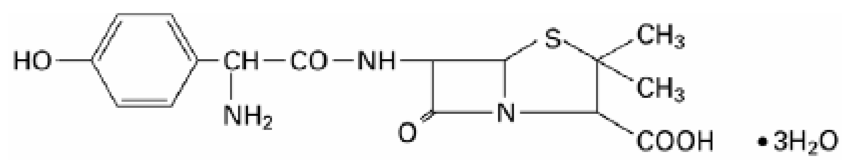
*Excipients*: Microcrystalline Cellulose, Sodium Starch Glycolate, Colloidal Silicon dioxide, Povidone (K30), Eudragit E 100, Isopropyl Alcohol, Magnesium stearate, Opadry 03B58965 white, Polyethylene glycol-400, Methylene chloride

## PRODUCT DESCRIPTION

White to off-white film coated oval shaped tablets debossed with "RX713" on one side and plain on the other

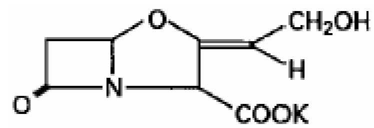
## DESCRIPTION

ENHANCIN TABLETS contain amoxicillin and clavulanate potassium. Amoxicillin/clavulanate potassium is an oral antibacterial combination consisting of the semisynthetic antibiotic amoxicillin and the β-lactamase inhibitor, clavulanate potassium (the potassium salt of clavulanic acid). Amoxicillin is an analog of ampicillin, derived from the basic penicillin nucleus, 6-aminopenicillanic acid. The amoxicillin molecular formula is $C_{16}H_{19}N_3O_5S \cdot 3H_2O$, and the molecular weight is 419.46. Chemically, amoxicillin is (2*S,* 5*R,* 6*R*)-6-[(*R*)-(-)-2-Amino-2-(*p*-hydroxyphenyl) acetamido]-3,3-dimethyl-7-oxo-4-thia-1-azabicyclo[3.2.0]heptane-2-carboxylic acid trihydrate and may be represented structurally as:




AMOXICILLIN

Clavulanic acid is produced by the fermentation of *Streptomyces clavuligerus*. It is a β-lactam structurally related to the penicillins and possesses the ability to inactivate a wide variety of β-lactamases by blocking the active sites of these enzymes. Clavulanic acid is particularly active against the clinically important plasmid-mediated β-lactamases frequently responsible for transferred drug resistance to penicillin's and cephalosporins. The clavulanate potassium molecular formula is $C_8H_8KNO_5$ and the molecular weight is 237.25. Chemically, clavulanate potassium is potassium (*Z*)-(2*R,* 5*R*)-3-(2-hydroxyethylidene)-7-oxo-4-oxa-1-azabicyclo[3.2.0]-heptane-2-carboxylate and may be represented structurally as:




CLAVULANATE POTASSIUM

PHARMACODYNAMIC AND PHARMACOKINETIC PROPERTIES

- Pharmacodynamics

Mechanism of action

Amoxicillin is a semisynthetic penicillin (beta-lactam antibiotic) that inhibits one or more enzymes (often referred to as penicillin-binding proteins, PBPs) in the biosynthetic pathway of bacterial peptidoglycan, which is an integral structural component of the bacterial cell wall. Inhibition of peptidoglycan synthesis leads to weakening of the cell wall, which is usually followed by cell lysis and death.

Amoxicillin is susceptible to degradation by beta-lactamases produced by resistant bacteria and therefore the spectrum of activity of amoxicillin alone does not include organisms which produce these enzymes.

Clavulanic acid is a beta-lactam structurally related to penicillins. It inactivates some beta-lactamase enzymes thereby preventing inactivation of amoxicillin. Clavulanic acid alone does not exert a clinically useful antibacterial effect.

Commonly susceptible species

Aerobic Gram-positive micro-organisms
*Enterococcus faecalis*
*Gardnerella vaginalis*
*Staphylococcus aureus (* methicillin-susceptible)[£]
Coagulase-negative staphylococci (methicillin-susceptible)
*Streptococcus agalactiae*
*Streptococcus pneumoniae*[1]
*Streptococcus pyogenes* and other beta-haemolytic streptococci

*Streptococcus viridans* group
[£]All methicillin-resistant staphylococci are resistant to amoxicillin and clavulanic acid.
[1]*Streptococcus pneumoniae* that are resistant to penicillin should not be treated with this presentation of amoxicillin/clavulanic acid.

<u>Aerobic Gram-negative micro-organisms</u>
*Capnocytophaga* spp.
*Eikenella corrodens*
*Haemophilus influenzae*[2]
*Moraxella catarrhalis*
*Pasteurella multocida*
[2]Strains with decreased susceptibility have been reported in some countries in the EU with a frequency higher than 10%.

<u>Anaerobic micro-organisms</u>
*Bacteroides fragilis*
*Fusobacterium nucleatum*
*Prevotella* spp.

Species for which acquired resistance may be a problem
<u>Aerobic Gram-positive micro-organisms</u>
*Enterococcus faecium* [$]
[$] Natural intermediate susceptibility in the absence of acquired mechanism of resistance.

<u>Aerobic Gram-negative micro-organisms</u>
*Escherichia coli*
*Klebsiella oxytoca*
*Klebsiella pneumoniae*
*Proteus mirabilis*
*Proteus vulgaris*

Inherently resistant organisms
<u>Aerobic Gram-negative micro-organisms</u>
*Acinetobacter* sp.
*Citrobacter freundii*
*Enterobacter* sp.
*Legionella pneumophila*
*Morganella morganii*
*Providencia* spp.
*Pseudomonas* sp.
*Serratia* sp.
*Stenotrophomonas maltophilia*

<u>Other micro-organisms</u>
*Chlamydophila pneumoniae*
*Chlamydophila psittaci*
*Coxiella burnetti*

*Mycoplasma pneumonia*

- Pharmacokinetics

Amoxicillin and clavulanic acid, are fully dissociated in aqueous solution at physiological pH. Both components are rapidly and well absorbed by the oral route of administration. Absorption of amoxicillin/clavulanic acid is optimised when taken at the start of a meal. Following oral administration, amoxicillin and clavulanic acid are approximately 70% bioavailable. The plasma profiles of both components are similar and the time to peak plasma concentration ($T_{max}$) in each case is approximately one hour.

About 25% of total plasma clavulanic acid and 18% of total plasma amoxicillin is bound to protein. The apparent volume of distribution is around 0.3-0.4 l/kg for amoxicillin and around 0.2 l/kg for clavulanic acid.

Following intravenous administration, both amoxicillin and clavulanic acid have been found in gall bladder, abdominal tissue, skin, fat, muscle tissues, synovial and peritoneal fluids, bile and pus. Amoxicillin does not adequately distribute into the cerebrospinal fluid.

There is no evidence for significant tissue retention of drug-derived material for either component. Amoxicillin, like most penicillins, can be detected in breast milk. Trace quantities of clavulanic acid can also be detected in breast milk.
Both amoxicillin and clavulanic acid have been reported to cross the placental barrier.

Amoxicillin is partly excreted in the urine as the inactive penicilloic acid in quantities equivalent to up to 10 to 25% of the initial dose. Clavulanic acid is extensively metabolized in man and eliminated in urine and faeces and as carbon dioxide in expired air.

The major route of elimination for amoxicillin is via the kidney, whereas for clavulanic acid it is by both renal and non-renal mechanisms.

Amoxicillin and clavulanic acid has a mean elimination half-life of approximately one hour and a mean total clearance of approximately 25 l/h has been reported. Approximately 60 to 70% of the amoxicillin and approximately 40 to 65% of the clavulanic acid are excreted unchanged in urine during the first 6 h after administration of single amoxicillin and clavulanic acid 250 mg/125 mg or 500 mg/125 mg tablets. The urinary excretion has been reported to be 50-85% for amoxicillin and between 27-60% for clavulanic acid over a 24 hour period. In the case of clavulanic acid, the largest amount of drug is excreted during the first 2 hours after administration.
Concomitant use of probenecid delays amoxicillin excretion but does not delay renal excretion of clavulanic acid.

*Age*

The elimination half-life of amoxicillin is similar for children aged around 3 months to 2 years and older children and adults. For very young children (including preterm newborns) in the first week of life the interval of administration should not exceed twice daily administration due to immaturity of the renal pathway of elimination. Because elderly patients are more likely to have decreased renal function, care should be taken in dose selection, and it may be useful to monitor renal function.

*Gender*

Following oral administration of amoxicillin and clavulanic acid, gender has no significant impact on the pharmacokinetics of either amoxicillin or clavulanic acid.

*Renal impairment*

The total serum clearance of amoxicillin and clavulanic acid decreases proportionately with decreasing renal function. The reduction in drug clearance is more pronounced for amoxicillin than for clavulanic acid, as a higher proportion of amoxicillin is excreted via the renal route. Doses in renal impairment must therefore prevent undue accumulation of amoxicillin while maintaining adequate levels of clavulanic acid.

*Hepatic impairment*

Hepatically impaired patients should be dosed with caution and hepatic function monitored at regular intervals.

## INDICATIONS

Amoxicillin + clavulanate potassium is an antibiotic agent with a notably broad spectrum of activity against the commonly occurring bacterial pathogens in general practice and hospital. The β-lactamase inhibitory action of clavulanate extends the spectrum of amoxicillin to embrace a wider range of organisms, including many resistant to other β-lactam antibiotics.

Amoxicillin + clavulanate potassium are indicated for short-term treatment of bacterial infections at the following sites:

*Upper respiratory tract infections (including ENT)* e.g. tonsillitis, sinusitis, otitis media.
*Lower respiratory tract infections* e.g. acute exacerbation of chronic bronchitis, lobar and bronchopneurnonia.
*Genito-urinary tract infections* e.g. cystitis, urethritis, pyelonephritis.
*Skin and soft tissue infections* e.g. boils, abscesses, cellulitis, wound infections.
*Bone and joint infections* e.g. osteomyelitis.
*Dental infections* e.g. dentoalveolar abscess
*Other infections* e.g. septic abortion, puerperal sepsis, intra-abdominal sepsis.

## DOSE AND METHOD OF ADMINISTRATION

ENHANCIN TABLETS is available as 625 mg strength which is not suitable for all dosage recommendations. Therefore, other suitable available strengths and dosage forms of amoxicillin and clavulanate potassium should be used in such cases.

*Adults*
The usual adult dose is one 625 mg tablet (Amoxicillin 500 mg + Clavulanate Potassium 125 mg) every 12 hours.

For more severe infections and infections of the respiratory tract, the dose should be one 625 mg tablet (Amoxicillin 500 mg + Clavulanate Potassium 125 mg) every 8 hours.

*Dosage in renal impairment*
Patients with impaired renal function do not generally require a reduction in dose unless the impairment is severe. Patients with a glomerular filtration rate of 10 to 30 mL/min. should receive one 625 mg tablet (Amoxicillin 500 mg + Clavulanate Potassium 125 mg) every 12 hours, depending on the severity of the infection. Patients with a less than 10 mL/min glomerular filtration rate should receive one 625 mg tablet (Amoxicillin 500 mg + Clavulanate Potassium 125 mg) every 24 hours, depending on severity of the infection.

Hemodialysis patients should receive one 625 mg tablet (Amoxicillin 500 mg + Clavulanate Potassium 125 mg) every 24 hours, depending on severity of the infection. They should receive an additional dose both during and at the end of dialysis.

*Dosage in hepatic impairment*
Hepatically impaired patients should be dosed with caution and hepatic function monitored at regular intervals.

*Pediatric patients weighing 40 kg or more*
In pediatric patients weighing 40 kg or more the dose should be one 625 mg tablet (Amoxicillin 500 mg + Clavulanate Potassium 125 mg) every 12 hours.

For more severe infections and infections of the respiratory tract, the dose should be one 625 mg tablet (Amoxicillin 500 mg + Clavulanate Potassium 125 mg) every 8 hours.

## Administration
Tablets should be swallowed whole without chewing.
To minimize potential gastrointestinal intolerance, administer at the start of a meal. The absorption of ENHANCIN TABLETS is optimized when taken at the start of a meal.

Treatment should not be extended beyond 14 days without review.

## CONTRAINDICATIONS
- Hypersensitivity to the amoxicillin/clavulanate potassium, or to any of the excipients.

- In patients with a history of hypersensitivity to beta-lactams, e.g. penicillins and cephalosporins.
- In patients with a previous history of amoxicillin/clavulanate potassium associated jaundice/hepatic dysfunction.

## WARNINGS AND PRECAUTIONS

SERIOUS AND OCCASIONALLY FATAL HYPERSENSITIVITY REACTIONS (INCLUDING ANAPHYLACTOID AND SEVERE CUTANEOUS ADVERSE REACTIONS) HAVE BEEN REPORTED IN PATIENTS RECEIVING THERAPY WITH BETA-LACTAMS. BEFORE INITIATING THERAPY WITH ENHANCIN TABLETS, CAREFUL INQUIRY SHOULD BE MADE CONCERNING PREVIOUS HYPERSENSITIVITY REACTIONS TO PENICILLINS, CEPHALOSPORINS, CARBAPENEMS OR OTHER BETA-LACTAM AGENTS. IF AN ALLERGIC REACTION OCCURS, ENHANCIN TABLETS MUST BE DISCONTINUED IMMEDIATELY AND APPROPRIATE ALTERNATIVE THERAPY INSTITUTED.

*Clostridium difficile* associated diarrhea (CDAD) has been reported with use of nearly all antibacterial agents, including amoxicillin/clavulanate potassium, and may range in severity from mild diarrhea to fatal colitis. Treatment with antibacterial agents alters the normal flora of the colon leading to overgrowth of *C. difficile*.

*C. difficile* produces toxins A and B which contribute to the development of CDAD. Hypertoxin producing strains of *C. difficile* cause increased morbidity and mortality, as these infections can be refractory to antimicrobial therapy and may require colectomy. CDAD must be considered in all patients who present with diarrhea following antibiotic use. Careful medical history is necessary since CDAD has been reported to occur over two months after the administration of antibacterial agents.

If CDAD is suspected or confirmed, ongoing antibiotic use not directed against *C. difficile* may need to be discontinued. Appropriate fluid and electrolyte management, protein supplementation, antibiotic treatment of *C. difficile* and surgical evaluation should be instituted as clinically indicated. Anti-peristaltic medicinal products are contra-indicated in this situation.

Amoxicillin/clavulanate potassium should be used with caution in patients with evidence of hepatic dysfunction. Hepatic events have been reported predominantly in males and elderly patients and may be associated with prolonged treatment. These events have been very rarely reported in children. In all populations, signs and symptoms usually occur during or shortly after treatment but in some cases may not become apparent until several weeks after treatment has ceased. These are usually reversible. Hepatic events may be severe and, in extremely rare circumstances, deaths have been reported. These have almost always occurred in patients with serious underlying disease or taking concomitant medications known to have the potential for hepatic effects.

Cholestatic jaundice, which may be severe, but is usually reversible, has been reported rarely. Signs and symptoms may not become apparent for up to six weeks after treatment has ceased.

The occurrence at the treatment initiation of a feverish generalized erythema associated with pustula may be a symptom of acute generalized exanthemous pustulosis (AGEP). This reaction requires amoxicillin/clavulanic potassium discontinuation and contra-indicates any subsequent administration of amoxicillin.

## Precautions

- General

While amoxicillin/clavulanate potassium possesses the characteristic low toxicity of the penicillin group of antibiotics, periodic assessment of organ system functions, including renal, hepatic, and hematopoietic function, is advisable during prolonged therapy.

Convulsions may occur in patients with impaired renal function or in those receiving high doses.

A high percentage of patients with mononucleosis who receive ampicillin develop an erythematous skin rash. Thus, ampicillin-class antibiotics should not be administered to patients with mononucleosis.

Prolongation of prothrombin time has been reported rarely in patients receiving amoxicillin/clavulanic acid. Appropriate monitoring should be undertaken when anticoagulants are prescribed concomitantly. Adjustments in the dose of oral anticoagulants may be necessary to maintain the desired level of anticoagulation.

The possibility of superinfections with mycotic or bacterial pathogens should be kept in mind during therapy. If superinfections occur (usually involving *Pseudomonas* or *Candida*), the drug should be discontinued and/or appropriate therapy instituted.

Prescribing amoxicillin/clavulanate potassium in the absence of a proven or strongly suspected bacterial infection or a prophylactic indication is unlikely to provide benefit to the patient and increases the risk of the development of drug-resistant bacteria.

In patients with reduced urine output, crystalluria has been observed very rarely, predominantly with parenteral therapy. During the administration of high doses of amoxicillin, it is advisable to maintain adequate fluid intake and urinary output in order to reduce the possibility of amoxicillin crystalluria. In patients with bladder catheters, a regular check of patency should be maintained.

### Geriatrics

No differences in responses between the elderly and younger patients have been reported, but a greater sensitivity of some older individuals cannot be ruled out.

This drug is known to be substantially excreted by the kidney, and the risk of toxic reactions to this drug may be greater in patients with impaired renal function. Because elderly patients are more likely to have decreased renal function, care should be taken in dose selection, and it may be useful to monitor renal function.

### Effect on Driving/Machine operating ability

No studies on the effects on the ability to drive and use machines have been reported. However, undesirable effects may occur (e.g. allergic reactions, dizziness, convulsions), which may influence the ability to drive and use machines.

### Information for patients

Patients should be counseled that antibacterial drugs including amoxicillin/clavulanate potassium, should only be used to treat bacterial infections. They do not treat viral infections (e.g., the common cold). When amoxicillin/clavulanate potassium is prescribed to treat a bacterial infection, patients should be told that although it is common to feel better early in the course of therapy, the medication should be taken exactly as directed.

Skipping doses or not completing the full course of therapy may: (1) decrease the effectiveness of the immediate treatment and (2) increase the likelihood that bacteria will develop resistance and will not be treatable by amoxicillin/clavulanate potassium or other antibacterial drugs in the future.

Diarrhea is a common problem caused by antibiotics which usually ends when the antibiotic is discontinued. Sometimes after starting treatment with antibiotics, patients can develop watery and bloody stools (with or without stomach cramps and fever) even as late as two or more months after having taken the last dose of the antibiotic. If this occurs, patients should contact their physician as soon as possible.

## DRUG INTERACTIONS

*Probenecid*

Probenecid decreases the renal tubular secretion of amoxicillin. Concurrent use with amoxicillin/clavulanate potassium may result in increased and prolonged blood levels of amoxicillin. Co-administration of probenecid cannot be recommended.

*Oral Anticoagulants*

Abnormal prolongation of prothrombin time (increased international normalized ratio [INR]) has been reported rarely in patients receiving amoxicillin and oral anticoagulants. Appropriate monitoring should be undertaken when anticoagulants are prescribed concurrently. Adjustments in the dose of oral anticoagulants may be necessary to maintain the desired level of anticoagulation.

*Allopurinol*

The concurrent administration of allopurinol and ampicillin increases substantially the incidence of rashes in patients receiving both drugs as compared to patients receiving ampicillin alone. It is not known whether this potentiation of ampicillin rashes is due to allopurinol or the hyperuricemia present in these patients. There are no information with amoxicillin and clavulanate potassium and allopurinol administered concurrently.

*Oral Contraceptives*

In common with other antibiotics, amoxicillin and clavulanate may affect the gut flora, leading to lower oestrogen reabsorption and reduced efficacy of combined oral contraceptives.

*Methotrexate*
Penicillins may reduce the excretion of methotrexate causing a potential increase in toxicity.

- Laboratory Test Interactions

During treatment with amoxicillin, enzymatic glucose oxidase methods should be used whenever testing for the presence of glucose in urine because false positive results may occur with non-enzymatic methods.

The presence of clavulanic acid in amoxicillin/clavulanate potassium may cause a non-specific binding of IgG and albumin by red cell membranes leading to a false positive Coombs test.

There have been reports of positive test results using the *Aspergillus* EIA test in patients receiving amoxicillin/clavulanic acid who were subsequently found to be free of *Aspergillus* infection. Cross-reactions with non-*Aspergillus* polysaccharides and polyfuranoses with *Aspergillus* EIA test have been reported. Therefore, positive test results in patients receiving amoxicillin/clavulanic acid should be interpreted cautiously and confirmed by other diagnostic methods.

## SIDE EFFECTS/UNDESIRABLE EFFECTS

### Infections and infestations
*Common*: Mucocutaneous candidiasis
*Not known*: Overgrowth of non-susceptible organisms

### Blood and lymphatic system disorders
*Rare*: Reversible leucopenia (including neutropenia), thrombocytopenia
*Not known*: Reversible agranulocytosis, haemolytic anaemia, prolongation of bleeding time and prothrombin time

### Immune system disorders
*Not known*: Angioneurotic oedema, anaphylaxis, serum sickness-like syndrome, hypersensitivity vasculitis

### Nervous system disorders
*Uncommon*: Dizziness, headache
*Not known*: Reversible hyperactivity, convulsions

### Gastrointestinal disorders
*Very common*: Diarrhea
*Common*: Nausea, vomiting
*Uncommon*: Indigestion
*Not known*: Antibiotic-associated colitis, black hairy tongue

### Hepatobiliary disorders
*Uncommon*: Rises in AST and/or ALT
*Not known*: Hepatitis, cholestatic jaundice

### Skin and subcutaneous tissue disorders

*Uncommon*: Skin rash, pruritus, urticaria
*Rare*: Erythema multiforme
*Not known*: Stevens-Johnson syndrome, toxic epidermal necrolysis, bullous exfoliative-dermatitis, acute generalised exanthemous pustulosis (AGEP)
Frequency 'very rare': Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS)

### Renal and urinary disorders

*Not known*: Interstitial nephritis, crystalluria

Hepatic events have been reported predominantly in males and elderly patients and may be associated with prolonged treatment. These events have been very rarely reported in children.
Signs and symptoms usually occur during or shortly after treatment but in some cases may not become apparent until several weeks after treatment has ceased. These are usually reversible. Hepatic events may be severe and in extremely rare circumstances, deaths have been reported. These have almost always occurred in patients with serious underlying disease or taking concomitant medications known to have the potential for hepatic effects.

The following adverse reactions have been reported for ampicillin-class antibiotics:
Agitation, anxiety, behavioral changes, confusion, insomnia, thrombocytopenic purpura, eosinophilia, hematuria, vaginitis, increases in serum bilirubin, and/or alkaline phosphatase.

Miscellaneous: Tooth discoloration (brown, yellow, or gray staining) has been rarely reported. Most reports occurred in pediatric patients. Discoloration was reduced or eliminated with brushing or dental cleaning in most cases.

## USE IN SPECIAL POPULATIONS

### Pregnancy

No teratogenic effects have been reported in animals (mice and rats) with orally and parenterally administered amoxicillin and clavulanate potassium.

It has been reported that prophylactic treatment with amoxicillin and clavulanate potassium in women with preterm, premature rupture of the foetal membrane (pPROM), may be associated with an increased risk of necrotising enterocolitis in neonates. As with all medicines, use should be avoided in pregnancy, especially during the first trimester, unless considered essential by the physician.

### Lactation

Both substances are excreted into breast milk (nothing is known of the effects of clavulanic acid on the breast-fed infant). Consequently, diarrhoea and fungus infection of the mucous membranes are possible in the breast-fed infant, so that breast-feeding might have to be discontinued. Amoxicillin and clavulanic acid should only be used during breast-feeding after benefit/risk assessment by the physician in charge.

## OVERDOSE

### Symptoms and signs of overdose

Gastrointestinal symptoms and disturbance of the fluid and electrolyte balances may be evident. Amoxicillin crystalluria, in some cases leading to renal failure, has been reported.

Convulsions may occur in patients with impaired renal function or in those receiving high doses.

Amoxicillin has been reported to precipitate in bladder catheters, predominantly after intravenous administration of large doses. A regular check of patency should be maintained.

### Treatment of intoxication

Gastrointestinal symptoms may be treated symptomatically, with attention to the water/electrolyte balance.

Amoxicillin and clavulanic acid can be removed from the circulation by haemodialysis.

## STORAGE:

Store below 30°C, protected from moisture.

Keep out of the reach and sight of children.

## SUPPLY

Blister pouch of 10 x 10's/box

MADE IN INDIA
Sun Pharmaceutical Ind. Ltd.
Industrial Area - 3
Dewas- 455001, India

PRODUCT REGISTRATION HOLDER:
RANBAXY (MALAYSIA) SDN. BHD.
(A Sun Pharma Company)
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08000 Sungai Petani, Kedah,
MALAYSIA

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